## LEGEND

TIE–IN POINT

——— – DARK LINES AND TEXT REPRESENT NEW EQUIPMENT, PIPING, AND INSTRUMENTATION.

——— – LIGHT LINES AND TEXT REPRESENT EXISTING, EXISTING TO BE RELOCATED, OR REFERENCE INFORMATION.

## SEQUENCE OF OPERATIONS

1) INCOMING WASTE FLOWS FREELY INTO THE TREATMENT TANK T–2.1.
2) FLUID RECIRCULATES THROUGH ENCLOSED PLUMBING, BEYOND THE pH SENSOR THROUGH V2.3 AND BACK INTO THE TANK VIA THE EDUCTION MIXING SYSTEM.
3) IF pH DEVIATES FROM THE USER DEFINED SETPOINTS THEN THE APPROPRIATE REAGENT IS INJECTED INTO THE RECIRCULATION LOOP FOR TREATMENT.
4) IF pH IS WITHIN ACCEPTABLE DISCHARGE LIMITS AND IS STABLE FOR A MINIMUM PERIOD OF TIME THEN CONTROL VALVE V2.3 CHANGES POSITION AND ALLOWS TANK CONTENTS TO FLOW TO DRAIN.
5) WHEN FLUID LEVEL REACHES LEH2–2.1 THEN THE INFLUENT FLOW IS INHIBITED UNTIL THE LEVEL DESCENDS BELOW LEH2–2.1.
6) IF FLUID LEVEL REACHES LEHH–2.1 THEN THE INFLUENT IS SHUTDOWN AS WITH LEH2–2.1 AND AN ALARM STATE IS ANNUNCIATED.
7) AS FLUID LEVEL DESCENDS BELOW LEL–2.1 THE TRANSFER PUMP IS SHUTDOWN AND THE METERING PUMPS ARE DISABLED.

## GENERAL NOTES

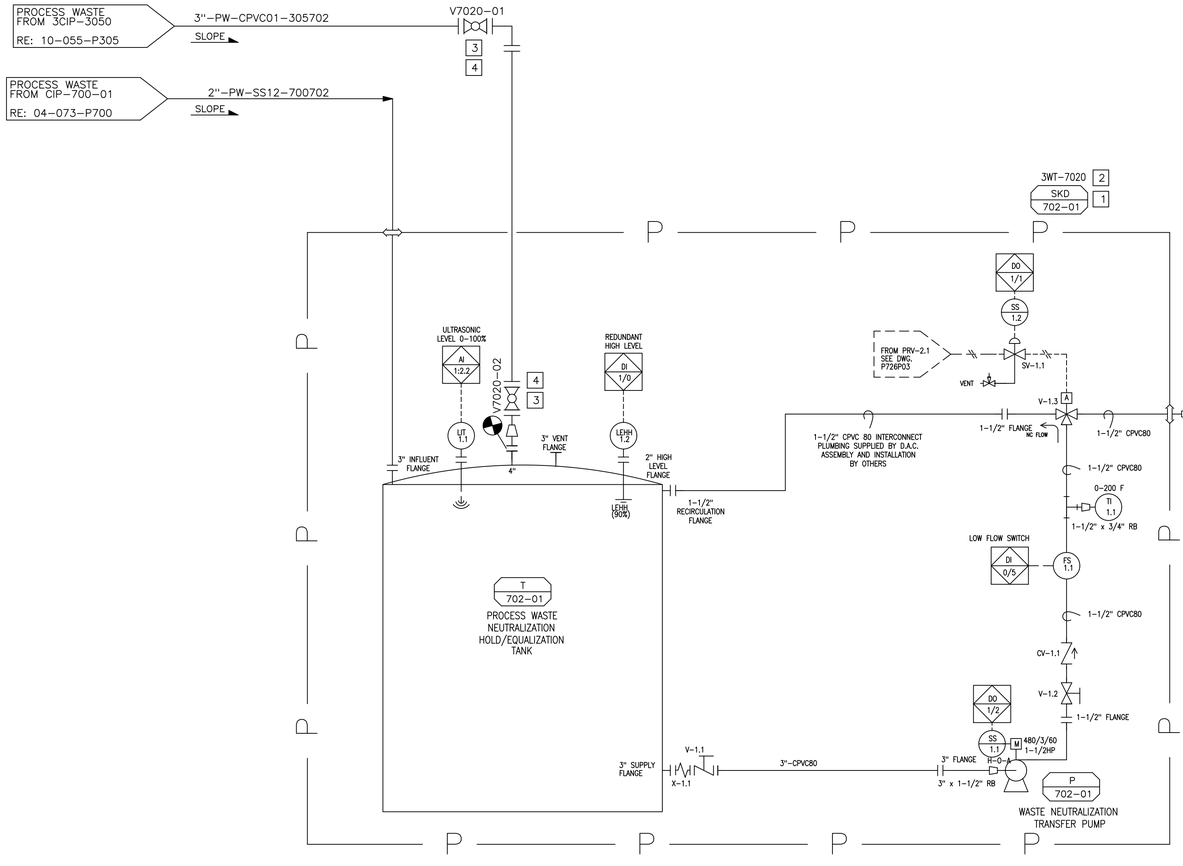
1. ALL NOTES ON DRAWING 10–055–G000 APPLY TO THIS DRAWING.

## P&ID NOTES

1. CONTROLS INCLUDED WITHIN PACKAGE.
2. PREFIX FOR SIGMA INSTRUMENT AND VALVE NUMBERS IS 3WT–7020– UNLESS NOTED OTHERWISE.
3. REMOVABLE SPOOL PIECE FOR ACCESS TO 4" INSPECTION FLANGE AT TOP HEAD OF TANK.
4. PROVIDE FLANGE COVERS ON ALL WASTE PIPING FLANGES.

## INTERLOCKS

| | | |
|---|---|---|
| SILENCE | DI 0/8 | TREATMENT SYSTEM AUDIBLE ALARM SILENCE PB |
| ALARM1 | DO 0/3 | PLC OUTPUTS RESERVED FOR TREATMENT SYSTEM FIELD PROGRAMMABLE ALARMS |
| FILL INHIBIT | DO 0/5 | |
| HORN | DO 1/0 | TREATMENT SYSTEM AUDIBLE ALARM / HORN |




LAST INSTR #: N/A
LAST VALVE #: 02

THIS RECORD DRAWING ISSUE (REVISION #5) HAS BEEN DRAFTED BY CE&IC BASED ON CONTRACTOR RED–LINED MARKUPS PROVIDED BY THE OWNER.

J
ISSUE DATE 05/26/15

| NO | REVISIONS | APPROVED | DATE |
|---|---|---|---|
| 5 | FOR RECORD ECM #500000002491 | JH | 05/22/15 |
| 4 | REVISED FOR CLIENT CHANGES | JDH | 02/04/15 |
| 3 | ISSUE FOR BID – N3 ADC PROJECT | SZ | 06/11/14 |
| 2 | ISSUE FOR BID | JDH | 04/04/14 |
| 1 | RECORD DRAWING | TCG | 04/24/07 |
| 0 | ISSUED FOR ADDENDUM #5 | LEM | 09/29/06 |

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ENGINEERING CERTIFICATE OF AUTHORITY NO. F01363794

SEAL

PROJECT LOCATION

SIGMA–ALDRICH
BUILDING N
3300 S. SECOND STREET
SAINT LOUIS, MO
63118
USA

SIGMA-ALDRICH
MANUFACTURING LLC

TITLE: ANIMAL PURIFICATION & NON–ANIMAL EXTRACTION & PURIFICATION WASTE NEUTRALIZATION SYSTEM PIPING & INSTRUMENTATION DIAGRAM

| | | | | | |
|---|---|---|---|---|---|
| PLANT: | CHEROKEE | BLDG: | N | FLOOR: | 1 |
| DRAWN: | MDW | SCALE: | NONE | DATE: | 09/13/06 |
| CHECKED: | | APPROVED: | | DATE: | – |

04–073–P702

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